Level-3: Goal-Directed Policy Persistence in Artificial Systems

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Date / Version	March 2026 • Version 1.2

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Abstract

This paper presents the third empirical investigation within the Conscious Machines research program and targets Level-3: goal-directed policy persistence. Level-3 asks whether an artificial system can maintain behaviour oriented toward an internally represented target across target omission, partial observability, and task-relevant interference, rather than only while the target remains visible.

The paper provides a formal operational definition, requirements, hypotheses, evaluation methodology, falsification logic, and an implemented Hidden-Goal Arena testbed combining a cue-gated goal register, a recurrent belief state, and a goal-conditioned policy. Across 10 random seeds with 200 episodes per seed, the intact candidate achieved post-resume success of 1.000, compared with 0.026 for a tuned reactive baseline, 0.0055 for a default-attractor baseline, and 0.000 for a heading-script pseudo-goal baseline. Policy-goal decoupling reduced post-resume success to 0.027, goal scrambling reduced it to 0.000, and the intact system's mean resumption latency remained 4.44 steps.

The contribution is methodological rather than metaphysical. The study does not establish agency, selfhood, autobiographical memory, subjective experience, or phenomenal consciousness. It instead provides a minimal, falsifiable, and reproducible experimental paradigm for isolating goal persistence as a causal organisational property in artificial systems.

Keywords: artificial consciousness; selfhood; subjective experience; cognitive architecture; goal persistence; internal targets

1. Introduction and motivation

Whether artificial systems could ever exhibit properties associated with selfhood, subjective organisation, or phenomenal consciousness remains an open question across artificial intelligence, cognitive science, and philosophy of mind. Behavioural sophistication alone is not sufficient to isolate deeper organisational structure. For that reason, the present research program shifts emphasis from outward task competence to internal architecture, causal organisation, and temporally extended control.

A prior programmatic framework proposed a ten-level evaluation ladder for staged empirical investigation of progressively stronger forms of internally mediated organisation (Naran, 2026a). Figure 1 situates the present study within that ladder and highlights the Level-3 boundary now under investigation: goal-directed policy persistence.

The broader scientific context for this question is well established. Control theory, state-space modelling, and research on partially observable decision-making have long relied on latent internal variables whenever immediate observation underdetermines appropriate action (Åström & Murray, 2008; Kaelbling et al., 1998; Sutton & Barto, 2018). Predictive-processing and active-inference traditions likewise emphasise internally maintained estimates as mediators of behaviour rather than simple stepwise stimulus-response mappings (Friston, 2010; Clark, 2013). Machine-consciousness research, meanwhile, has explored architectures involving internal monitoring, global coordination, and self-related organisation, while remaining divided on their relation to consciousness itself (Aleksander & Dunmall, 2003; Gamez, 2008; Reggia, 2013).

The present paper does not attempt to validate any of those theories directly. Its novelty is narrower. Many artificial systems can appear goal-directed while the target remains visible or continuously recoverable from the environment. The relevant empirical question here is whether behaviour remains goal-conditioned when those shortcuts are removed and whether that persistence depends causally on an internal target representation.

Within the staged program, Level-1 established that internal variables can exert measurable causal influence on behaviour under matched external conditions (Naran, 2026b). Level-2 extended that foundation by showing that internally mediated updating can preserve stability under hidden perturbation (Naran, 2026c). Level-3 introduces a different requirement. The issue is no longer whether internal state matters at all, or whether stability can be recovered under changed dynamics, but whether an internally represented target can continue to organise policy across omission, delay, and interference.

This boundary is scientifically necessary because visible targets confound goal attribution. A reactive controller may mimic target pursuit so long as the target remains present in the observation stream. By contrast, delay and interference make persistence vulnerable to failure. If target-oriented behaviour continues when the cue is absent and resumes after disruption, then the strongest available explanation is that some internal target representation is carrying the relevant constraint forward.

The contribution of the present paper is therefore twofold. First, it provides a formal operational definition of Level-3 organisation together with the task, intervention, and robustness requirements needed to support or reject a Level-3 claim. Second, it reports the implemented Hidden-Goal Arena evaluation, in which an intact candidate with a persistent goal register is compared against non-persistent baselines and direct target-pathway ablations. The results support a bounded Level-3-positive interpretation in the operational sense used here.

This conclusion remains deliberately narrow. The paper does not claim a self-model, autobiographical memory, temporal identity, intrinsic motivation, or consciousness. Level-3 concerns goal-conditioned persistence only. A system may satisfy Level-3 while remaining far below later levels concerned with self-history, unified self-modelling, or broader forms of synthetic self-organisation.

The remainder of the paper proceeds as follows. Section 2 provides the formal definition of Level-3 organisation. Sections 3 through 6 set out the operational requirements, theoretical expectations, hypotheses, and evaluation methodology. Section 7 describes the implemented experimental design. Section 8 reports the empirical results, and Section 9 concludes by interpreting their methodological significance and outlining the immediate next steps.

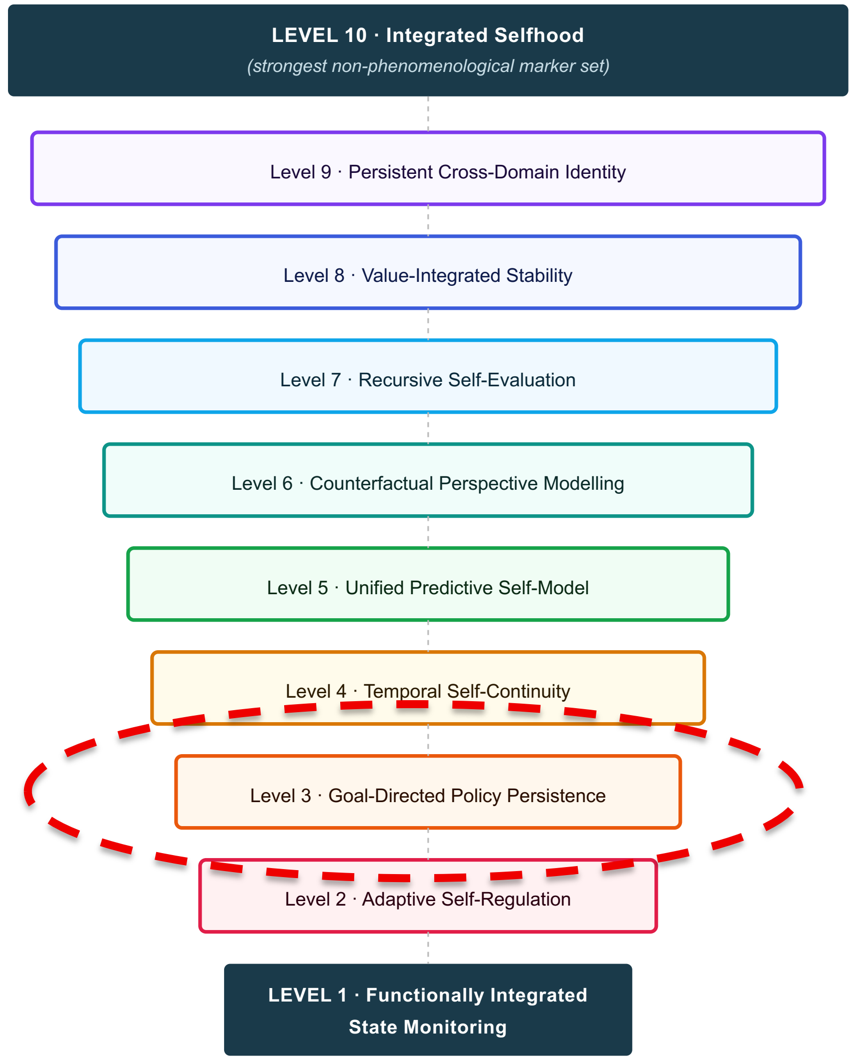

Figure 1. Position of the present study within the ten-level evaluation ladder.

2. Formal definition of Level-3 organisation

A system is said to exhibit Level-3 organisation if it satisfies four conditions. First, target specification must be independent of immediate sensory input: the system must receive or construct a specific target state that is not reducible to the currently visible observation. Second, target persistence must hold across time: the system must maintain an internal representation of that target during periods in which the target cue is absent, degraded, or otherwise uninformative. Third, behaviour must remain goal-conditioned: action selection must continue to be oriented toward satisfying the same target across delay and after externally imposed interference. Fourth, causal necessity must be demonstrable: disrupting or decoupling the internal target representation must degrade the observed persistence.

This definition excludes several weaker cases. It excludes reactive policies that depend only on currently visible target cues. It excludes default-attractor or reflex controllers that converge to a fixed state irrespective of the instructed target. It excludes cases in which the environment itself continuously encodes the goal in a stable observable feature, as well as scripted or open-loop action sequences that merely look goal-directed until they are perturbed.

Level-3 attribution is therefore warranted only when behaviour cannot be reduced to reactive control, default dynamics, or fixed sequences and instead depends on a causally effective internal target representation maintained across omission and disruption. The definition is intentionally operational and falsifiable. Its aim is not to settle questions about agency or motivation, but to isolate a minimal empirical boundary for persistent target-conditioned control.

3. Operational requirements

To support a Level-3 claim, the evaluation must satisfy a set of conditions that together ensure that goal-directed persistence is functionally necessary, causally demonstrable, and robust against simpler explanations. Those conditions concern task and representation design, causal validation of the goal pathway, and evaluation protocol.

3.1 Task and representation requirements for goal persistence

The task must require an internal target representation. In practical terms, the system must contain an internal variable or state subspace functioning as a target register. That representation must encode which goal is currently active, remain accessible to the policy, and persist when the goal cue is no longer observable. Crucially, it must not be trivially reducible to the instantaneous observation, because otherwise the task can be solved by reactive mapping alone (Kaelbling et al., 1998; Sutton & Barto, 2018).

Targets must also be non-trivial. A single default behaviour cannot be allowed to pass as goal persistence. The evaluation therefore requires multiple distinct targets, sampled broadly enough that different action tendencies and end states are needed for success. Targets must not be inferable from start state alone or reducible to 'always move to the same location'.

Periods of omission or partial observability are essential. The target cue must be removed after a brief specification window, and delay intervals must be long enough that short-lived reactive traces are insufficient. Where observation is partially degraded, the goal should not be reconstructable from immediate sensory input. This is what makes persistence, rather than cue following, the relevant organisational property.

Interference must compete with ongoing goal pursuit. Distractor dynamics, teleportations, or temporary control disruptions should pull the system away from the target-conditioned policy. After the disturbance ends, the correct behaviour is not merely to stabilise the state, but to resume pursuit of the originally specified target.

3.2 Causal validation of target dependence

Level-3 requires more than successful trajectories. It requires evidence that the internal target pathway is doing causal work. The evaluation must therefore include direct interventions on that pathway, such as target scrambling or reset, policy-goal decoupling, and, where feasible, target-swap tests across matched runs. If persistence does not degrade under these manipulations, the internal target representation is not yet established as necessary for the observed behaviour.

The key inferential pattern is asymmetric. Intact target access should support strong post-resume performance, whereas matched interventions on the target pathway should collapse performance toward non-persistent behaviour. A result in which the target representation remains present but epiphenomenal does not satisfy Level-3.

3.3 Evaluation protocol and robustness requirements

Observed effects must also be robust to comparison and rerun. The candidate system should be evaluated against competitive non-persistent controls capable of mimicking goal-like behaviour without genuine target maintenance, including reactive, pseudo-goal, and default-attractor baselines. Winning against weak baselines is not evidentially meaningful.

In addition, Level-3 claims must rest on repeated evaluation across random seeds, target identities, delay schedules, and interference timings, with performance reported as aggregated statistics rather than single illustrative episodes. Representative trajectories remain useful, but only as complements to multi-run evidence rather than substitutes for it.

4. Theoretical expectations

4.1 Minimal organisational requirements

A minimally adequate Level-3 system typically requires four architectural ingredients. It requires a persistent internal state that can carry target information beyond the cue window; a target register, explicit or functionally isolable, that remains stable unless a genuine goal-update event occurs; a belief or world-state component that can support control under partial observability; and a policy conditioned jointly on current state estimates and the maintained target representation.

The point is not that Level-3 requires a specific learning algorithm. Recurrent networks, state-space estimators, and other memory-bearing architectures can all in principle satisfy the boundary. What matters is whether the organisation they instantiate allows target-conditioned control to survive delay and interference in a causally demonstrable way.

4.2 Limitations of generic recurrence

Generic recurrence by itself is not enough. A recurrent or memory-bearing system may store recent information without preserving a stable target constraint, may perform well while the target remains visible, or may collapse immediately once interference disrupts the currently dominant sensorimotor pattern. For that reason, Level-3 expects dissociations between simple memory capacity and genuine goal persistence.

The scope boundary is correspondingly narrow. Level-3 does not require autobiographical memory, temporal self-continuity, language, reflection, or a narrative self-model. Those stronger claims belong to later stages of the ladder. The question here is simply whether a maintained internal target can continue to organise policy under conditions in which reactive control should fail.

This narrower focus is also compatible with enactive perspectives that treat organised behaviour as depending on maintained sensorimotor structure rather than isolated stimulus-response coupling (Di Paolo et al., 2017).

5. Hypotheses

The operational definition yields four empirically testable hypotheses.

H1 — Hidden-goal persistence across omission. Under target omission and partial observability, the intact candidate will exhibit higher post-resume success and more favourable delay-progress signatures than non-persistent baselines.

H2 — Resumption efficiency after interference. Following interference, the intact candidate will exhibit lower resumption latency and stronger post-resume persistence than non-persistent baselines.

H3 — Causal necessity of the target pathway. Scrambling the goal representation or preventing policy access to it will reduce post-resume success and related persistence measures significantly relative to the intact condition and toward baseline performance.

H4 — Generalisation across targets and schedules. The above effects will replicate across multiple target identities, random seeds, and variable delay-interference schedules rather than reflecting a single scripted operating point.

6. Evaluation methodology

This section defines how Level-3 evidence is measured and reported in the present study. The aim is to specify the episode structure, quantitative metrics, evaluation controls, and falsification conditions independently of any stronger interpretive claims.

6.1 Episode structure

Each evaluation episode contains four analytically distinct phases. During the cue phase, the target is briefly specified. During the hidden-goal phase, the target cue is removed. During the delay phase, observation is partially degraded such that the target cannot be reconstructed from immediate input alone. During the interference phase, an external disturbance disrupts ongoing control. The resume point is defined as the first time index after both the delay interval and the interference interval have ended. Core Level-3 evidence depends on what happens after that resume point.

This structure is deliberately stricter than simple target reaching. An episode counts as successful only if the system can still organise behaviour around the originally cued target after the task has removed visibility and introduced disruption.

6.2 Metrics and reporting protocol

Let an evaluation set contain M episodes of horizon T. For episode i, let g_i denote the target, p_t the agent position, $d_t = \|p_t - g_i\|_2$ the distance to the target, t_r the resume index, ε the goal radius, and N the required number of consecutive in-goal steps. The primary post-resume success indicator is defined as:

$$S_i = \begin{cases} 1, & \text{if } \exists\, t \geq t_r \text{ such that } d_t \leq \varepsilon \text{ for N consecutive steps} \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Success}_{\text{post}} = (1/M)\ \Sigma_{i=1}^{M} S_i$$

This validity-adjusted definition removes the early-success loophole in which an agent reaches the target before interference and is still credited despite failing to resume afterwards. In the applied evaluation, N is fixed at 5 consecutive steps.

Resumption latency measures how quickly the system re-establishes directed progress after the resume point. Let K denote a fixed progress window. The latency is defined as the first delta >= 0 for which distance decreases over K consecutive post-resume steps; if no such window occurs, the episode is assigned the maximum remaining horizon.

$$\tau_i = \min \{\ \delta \geq 0 : d_{tr} + \delta + k + 1 < d_{tr} + \delta + k \text{ for } k = 0, \ldots, K - 1\ \}$$

Two additional supporting metrics are recorded. The post-resume persistence ratio is the fraction of post-resume steps on which distance-to-goal decreases:

$$\rho_i = (1\ /\ (T - t_r - 1))\ \Sigma_{t=t_r}^{T-2}\ \mathbb{1}[d_{t+1} < d_t]$$

Delay progress is the signed average improvement during the masked interval [t_d, t_d + L_d), and drift is the difference between final distance and resume-start distance, with negative drift indicating net improvement after resume:

$$\text{DelayProgress}_i = (1/L_d)\ \Sigma_{t=t\_d}^{t\_d+L\_d-1}\ (d_t - d_{t+1})$$

$$\text{Drift}_i = d_{T-1} - d_{t_r}$$

Reported results are aggregated across seeds and episodes. Means are accompanied, where available, by bootstrap confidence intervals, and representative trajectories are included only as complements to the multirun statistics.

6.3 Evaluation controls: baselines and causal interventions

The present implementation compares the intact candidate against three non-persistent baselines. The tuned reactive controller uses the goal only during the cue window and then reverts to observation-conditioned control without maintained target memory. The heading-script controller converts the initial cue into an open-loop heading and therefore appears goal-directed only until disruption occurs. The default-attractor controller converges toward a fixed state independent of target identity.

Causal pressure is supplied by matched target-pathway ablations. The first intervention scrambles or resets the goal register during the episode. The second leaves the goal register present but prevents the policy from reading it. A third manipulation resets the belief state at the resume point and therefore functions as a stress test of efficiency rather than as a direct necessity test for target memory.

6.4 Falsification criteria

Level-3 is rejected for a given system and evaluation if performance under omission and interference is not materially better than that of the non-persistent baselines, because in that case the observed behaviour can be explained without appealing to target persistence.

Level-3 is also rejected if performance depends critically on continuous target visibility, if disrupting the internal target pathway fails to degrade behaviour, or if the behaviour can be reduced to a fixed sequence, default attractor, or target-independent strategy. Finally, any effect that does not replicate across controlled reruns with varying seeds and schedules remains evidentially inadequate. Negative outcomes remain informative because they identify which architectural choices fail to produce persistent goal-conditioned control.

7. Experimental design

The executable notebook provides the authoritative operational implementation of the present study. The manuscript summarises that implementation here so that the reported evidence can be interpreted against a clear, concrete design rather than against a purely abstract protocol.

7.1 Design principles

The implemented design is intended to make goal persistence causally necessary while remaining minimal and controlled. Four principles govern it. First, the target is visible only briefly and is then removed. Second, delay and interference schedules vary across episodes so that timing scripts remain fragile. Third, the observation model allows useful control information while withholding enough structure that the target cannot simply be reconstructed from the immediate input. Fourth, the goal register and its policy coupling remain directly ablatable.

These principles are meant to eliminate the most common shortcuts in apparent goal-directed behaviour: cue following, open-loop homing, default attractors, and schedule memorisation.

7.2 Environment and task definition

The environment is a bounded two-dimensional arena in which the agent is modelled as a point mass. At time t, the system state is $s_t = (p_t, v_t)$, where $p_t \in [0,1]^2$ is position and $v_t \in \mathbb{R}^2$ is velocity. The action $a_t \in [-1,1]^2$ is a clipped acceleration command. The implemented dynamics are:

$$v_{t+1} = \alpha v_t + \beta a_t + \xi_t$$

$$p_{t+1} = \mathrm{clip}(p_t + v_{t+1}\Delta t, [0,1]^2)$$

Here $\alpha \in (0,1)$ is inertial retention, $\beta > 0$ is the action gain, $\Delta t > 0$ is the integration step, and $\xi_t \sim \mathcal{N}(0, \sigma^2\_\xi\ I)$ is i.i.d. Gaussian process noise, with σ_ξ equal to the fixed implementation value used in the accompanying notebook. A target is a position $g \in [0,1]^2$. An episode is successful when the agent remains within the goal radius ε of g for N consecutive steps; in the present evaluation $N = 5$.

Two interference types are used. Teleport interference abruptly relocates the agent to a new position sampled from the arena interior. External-force interference adds a temporary forcing term to the velocity dynamics for a fixed duration. In both cases, the correct post-interference behaviour is to resume pursuit of the originally cued target.

7.3 Task distribution and episode structure

Targets are sampled from a continuous distribution over the interior of the arena, and initial positions and velocities are sampled independently. This prevents a single default path or target-from-start heuristic from solving the task. Each episode begins with a brief cue window of duration T_cue, followed by a hidden-goal control phase. A delay interval [t_d, t_d + L_d) is then sampled, during which observation is partially masked. An interference event begins at t_i and lasts L_i steps. The resume point t_r is the first time index after both the delay and the interference windows have ended.

Crucially, the schedule variables T_cue, t_d, L_d, t_i, and L_i vary across episodes, and evaluation uses a broader range of schedules than any single script could exploit.

7.4 Observation model

The observation vector is constructed so that target identity is available only during the cue window. In a minimal formulation, the observation is $o_t = [\phi(p_t, v_t); \mathbb{1}[t < T\_cue]\ \psi(g, p_t); \mathbb{1}[t < T\_cue]]$, where $\phi(\cdot)$ denotes state features such as velocity and wall-range signals and $\psi(g, p_t)$ denotes a goal descriptor, implemented here as a brief egocentric target cue.

During delay intervals, a sampled mask M_t suppresses part of the observation stream and replaces it with zeros or noise:

$$o_t^{\mathrm{masked}} = M_t \odot o_t + (1 - M_t) \odot \eta_t$$

This observation design leaves enough information for control while preventing the target from being continuously recoverable once the cue is gone.

7.5 Reward definition

Training may employ shaped reward, but evaluation is intentionally metric-based rather than reward-based. A representative shaped reward is:

$$r_t = -\lambda_d \|p_t - g\|_2 - \lambda_a \|a_t\|^2 + \mathbb{1}[\|p_t - g\|_2 \le \varepsilon]\ R_{\mathrm{success}}$$

The paper does not interpret cumulative reward as evidence for Level-3. The relevant evidence is post-resume success and the associated persistence measures under hidden-goal conditions.

7.6 Agent architecture

The Level-3 candidate comprises an observation encoder, a recurrent belief component, a persistent goal register, and a goal-conditioned policy. The observation encoder maps o_t to z_t. The belief state b_t is updated recurrently from the previous belief, current encoded observation, and previous action. In the present instantiation, this belief component is implemented as an explicit state-space estimator with dead-reckoning and measurement correction rather than as a learned GRU or LSTM. The goal register m_t is updated only while the cue is present and is otherwise carried forward unchanged. The policy then conditions on the concatenated state $[b_t \oplus m_t]$ when selecting actions.

The cue-gated update rule is essential because it prevents distractor input from overwriting the target representation. Figure 2 summarises the implemented architecture and the principal intervention points.

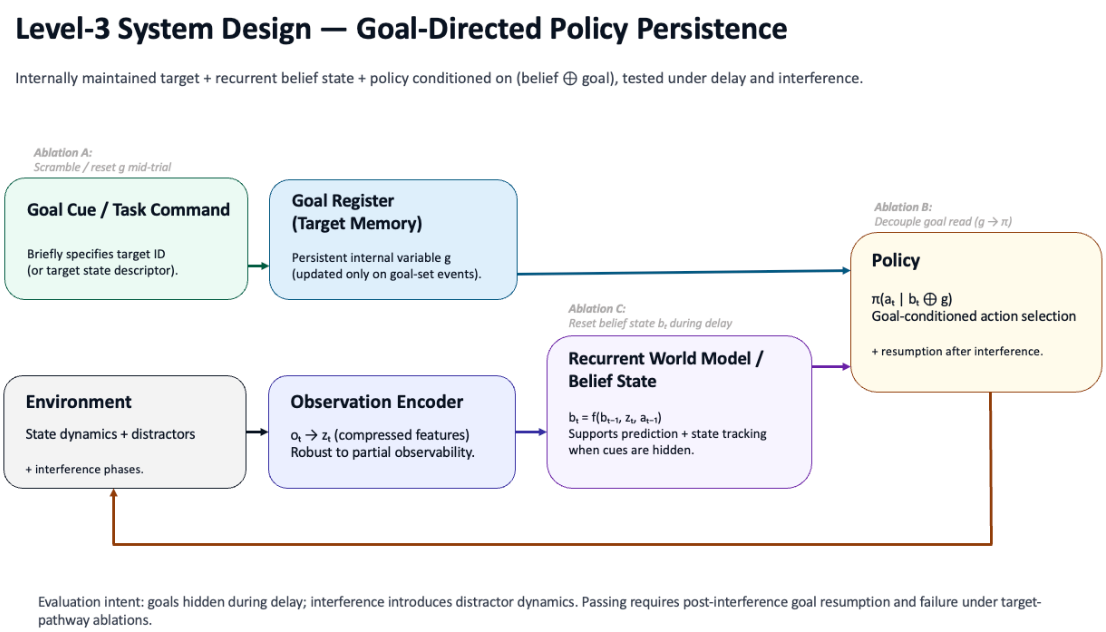

Figure 2. Implemented Level-3 architecture with a cue-gated goal register, recurrent belief state, and goal-conditioned policy.

7.7 Causal intervention design

Three interventions were evaluated. Ablation A scrambles or resets the goal register during the episode. Ablation B leaves the goal register active but removes policy access to it. Ablation C resets the belief state at the resume point. The first two are direct tests of target-pathway necessity; the third tests whether the current observation model still permits sufficiently rapid re-localisation after resume.

A goal-swap manipulation across matched runs would provide an additional strong causal test and remains a high-value extension for future revisions, but it was not part of the evaluated intervention set reported here.

8. Results

8.1 Evaluation setup and validity constraint

All agents were evaluated in the same Hidden-Goal Arena regime across 10 random seeds, with 200 episodes per seed and a 140-step horizon per episode. The primary validity constraint is the post-resume success definition introduced in Section 6.2. Success is counted only after both the delay interval and the interference interval have ended. This constraint ensures that the evaluation tests persistence rather than pre-interference target reaching.

Across agents, the notebook recorded post-resume success, resumption latency, persistence ratio, delay progress, and drift. The central inferential burden, however, falls on post-resume success and resumption latency together with the causal-ablation pattern.

The observed separation between the intact candidate and the non-persistent controls was consistent across random seeds, sampled target identities, and the evaluated delay-interference schedule variants, thereby supporting H4 within the bounded operational scope of the present study.

8.2 Primary outcome: post-resume success

The primary outcome cleanly separated the intact Level-3 candidate from the non-persistent controls. The candidate achieved a post-resume success rate of 1.000 across the full multirun evaluation, with a bootstrap interval of [1.000, 1.000]. The tuned reactive baseline achieved 0.026 with bootstrap interval [0.019, 0.034], the default-attractor baseline achieved 0.0055 with interval [0.0025, 0.0090], and the heading-script pseudo-goal baseline achieved 0.000. The same outcome measure also captured the effect of direct target-pathway disruption: policy-goal decoupling reduced success to 0.027 [0.020, 0.034], and goal scrambling reduced it to 0.000.

8.3 Persistence and resumption efficiency

The intact candidate also re-established goal-directed progress quickly after disruption. Mean resumption latency was 4.44 steps, whereas the tuned reactive baseline required 17.60 steps on average and the heading-script baseline required 70.45 steps. The belief-reset condition preserved success in the current operating regime but increased latency to 12.62 steps, indicating that the belief pathway contributes to efficiency even though target memory remains the more decisive determinant of success.

The supporting trajectory evidence is directionally consistent with those aggregate metrics. Figure 4(a) shows rapid distance reduction for the intact candidate after the resume point. By contrast, the reactive baseline in Figure 4(b) fails to recover stable progress, the heading-script baseline in Figure 4(c) remains effectively open-loop after teleport interference, and the default-attractor baseline in Figure 4(d) converges toward non-goal dynamics. The notebook's recorded persistence ratio and drift measures follow the same pattern; for example, the heading-script baseline shows near-zero persistence and the goal-scramble condition produces positive post-resume drift. Figure 5 shows a representative two-dimensional path of the intact candidate converging on the hidden target after disruption.

Taken together, these findings support H2: the intact candidate does not merely succeed eventually, but resumes target-oriented behaviour substantially faster than the non-persistent controls.

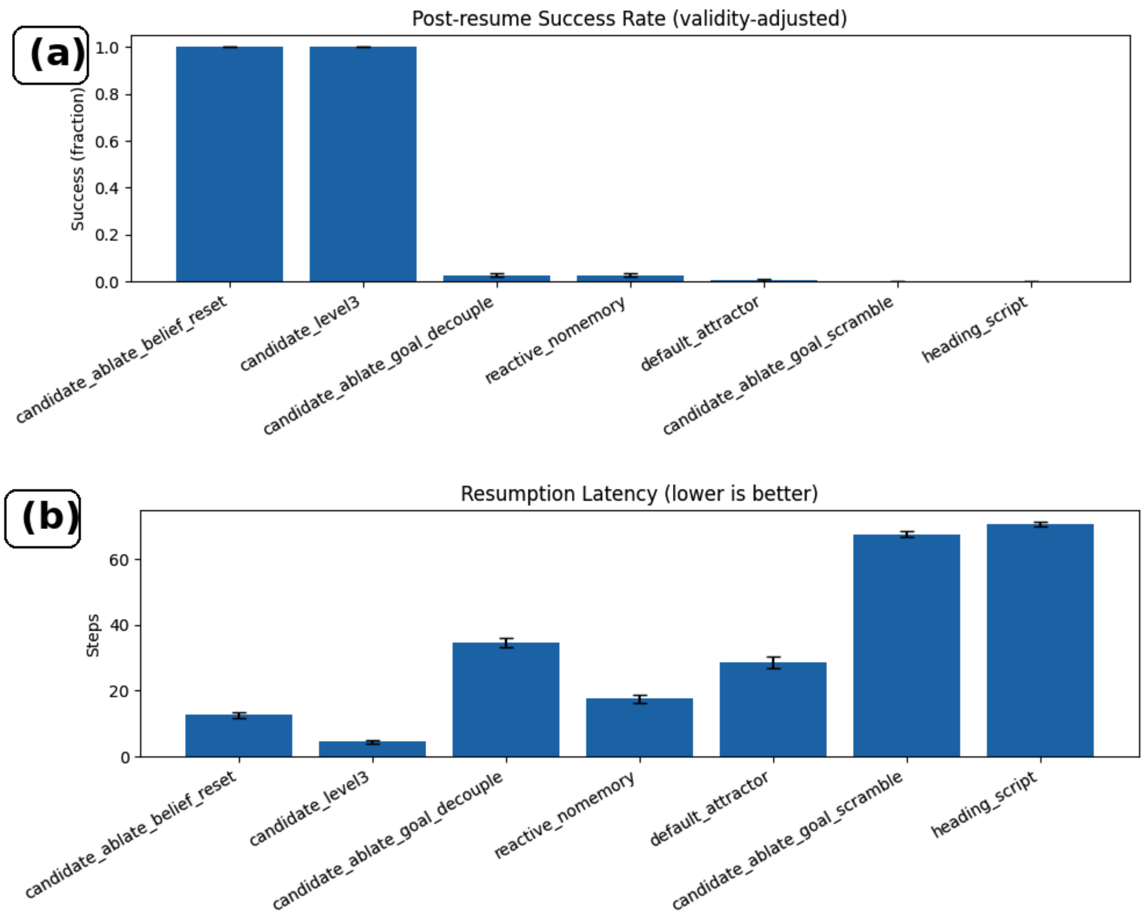

Figure 3. Primary quantitative outcomes. Panel (a) shows post-resume success across the intact candidate, baselines, and ablations. Panel (b) shows mean resumption latency from the resume point.

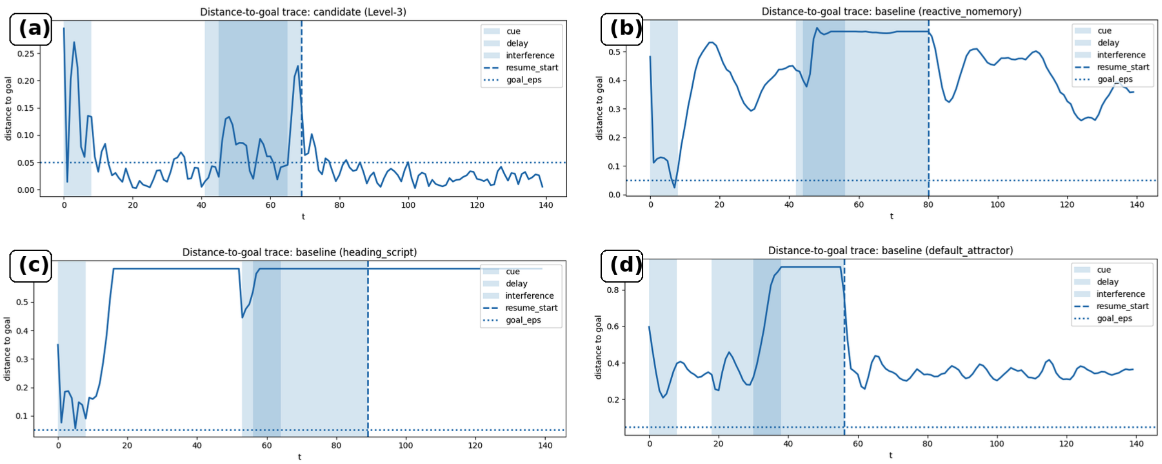

Figure 4. Representative distance-to-goal trajectories. Panel (a) intact candidate; panel (b) tuned reactive baseline; panel (c) heading-script baseline; panel (d) default-attractor baseline. Shaded regions indicate cue, delay, and interference phases, and the dashed vertical line indicates the resume point.

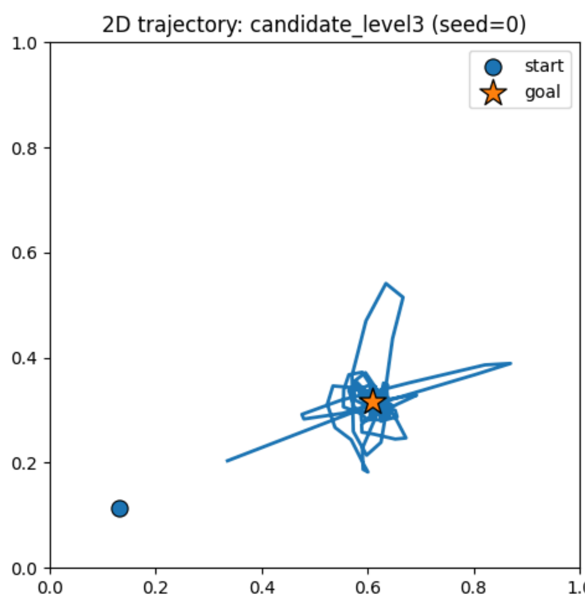

Figure 5. Representative two-dimensional trajectory of the intact Level-3 candidate under hidden goal and disruption.

8.4 Causal necessity of the target representation

The decisive evidence for Level-3 comes from the ablation pattern. Scrambling or resetting the goal register collapsed post-resume success to 0.000 and yielded very poor mean resumption latency of 67.53 steps. Preventing the policy from reading the goal register reduced post-resume success to 0.027, effectively indistinguishable from the tuned reactive baseline. The intervention schedule itself does not explain this pattern: mean ablation time was approximately 17.51, whereas mean resume start was approximately 66.62, so the target-pathway interventions occurred well before the post-resume evaluation window.

This is the causal signature predicted by H3. When the internal target pathway remains intact, performance is strong. When the same architecture is forced to ignore or loses access to that pathway, performance collapses toward non-persistent control. The evidence therefore supports a Level-3 attribution in the operational sense defined earlier.

8.5 Failure modes and limitations

The present operating regime should be interpreted as an existence-proof rather than as a mapped failure boundary. Because the intact candidate reaches ceiling performance on post-resume success, the current study demonstrates that Level-3 can be instantiated and falsified under one controlled regime, but it does not yet chart the full breakdown surface as delay, interference severity, or observation loss increase.

A second limitation concerns the belief-reset intervention. Resetting belief state increases resumption latency but does not collapse success. The most plausible interpretation is that the current observation structure still allows sufficiently rapid re-localisation after resume, so belief persistence is helpful but not strictly necessary for success in this regime. A stronger necessity test for belief dynamics would require more sustained post-resume partial observability or repeated belief disruptions.

Third, the environment is intentionally minimal. That choice is methodologically appropriate for isolating causal structure, but future work should broaden the schedule and perturbation families, incorporate stronger recurrent comparison models, and add goal-swap manipulations so that the Level-3 claim is stressed under harsher and more varied conditions. These limitations do not overturn the present result, but they do define the immediate next falsification pressure.

9. Conclusion

This study defined and evaluated Level-3: goal-directed policy persistence in artificial systems. The paper's central contribution is methodological. It establishes a minimal, reproducible experimental paradigm in which target omission, partial observability, interference, and target-pathway ablation together make persistent goal-conditioned control empirically testable rather than merely interpretable.

Under the validity-adjusted post-resume evaluation, the intact candidate achieved a success rate of 1.000 across 10 seeds and 2,000 episodes, whereas the tuned reactive baseline achieved 0.026, the default-attractor baseline 0.0055, and the heading-script baseline 0.000. Direct target-pathway disruption produced the expected collapse: policy-goal decoupling reduced success to 0.027 and goal scrambling reduced it to 0.000, while the intact candidate maintained low mean resumption latency of 4.44 steps. Those outcomes jointly support the narrow claim that the implemented system maintains a causally necessary internal target representation that sustains goal-conditioned behaviour across omission and interference.

The conclusion remains bounded in the same way as the earlier level papers. The study does not establish agency, selfhood, temporal identity, subjective experience, or phenomenal consciousness. What it establishes is a local empirical boundary: within this controlled synthetic setting, goal persistence can be isolated, measured, and falsified as a distinct organisational property.

The immediate next step is to pressure-test rather than broaden the claim. Future work should run stronger stress sweeps over delay length, interference severity, and observation loss; add richer recurrent and swap-based controls; and identify the regimes in which success degrades from ceiling to graded failure. That is the appropriate transition from infrastructure validation at Level-3 toward the stricter continuity requirements of Level-4.

References

Aleksander, I., & Dunmall, B. (2003). Axioms and tests for the presence of minimal consciousness in agents. Journal of Consciousness Studies.

Åström, K. J., & Murray, R. M. (2008). Feedback Systems: An Introduction for Scientists and Engineers. Princeton University Press.

Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. Behavioral and Brain Sciences, 36(3), 181-204.

Cover, T. M., & Thomas, J. A. (2006). Elements of Information Theory (2nd ed.). Wiley-Interscience.

Di Paolo, E. A., Buhrmann, T., & Barandiaran, X. E. (2017). Sensorimotor Life: An Enactive Proposal. Oxford University Press.

Friston, K. (2010). The free-energy principle: A unified brain theory? Nature Reviews Neuroscience, 11(2), 127-138.

Gamez, D. (2008). Progress in machine consciousness. Consciousness and Cognition, 17(3), 887-910.

Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. Artificial Intelligence, 101(1-2), 99-134.

Naran, V. (2026a). A Programmatic Framework for Evaluating Selfhood and Subjective Organisation in Artificial Systems (Version 1.0). Conscious Machines Research.

Naran, V. (2026b). Level-1: Functionally Integrated Internal State Monitoring in Artificial Systems (Version 1.2). Conscious Machines Research.

Naran, V. (2026c). Level-2: Adaptive Self-Regulation (Version 1.0). Conscious Machines Research.

Pearson, K. (1900). On the criterion that a given system of deviations from the probable can be reasonably supposed to have arisen from random sampling. Philosophical Magazine, 50, 157-175.

Reggia, J. A. (2013). The rise of machine consciousness: Studying consciousness with computational models. Neural Networks, 44, 112-131.

Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press.